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Next item →

1. What simplification is made to SPKF when process and sensor noises are both additive? (Select all that apply.)

1 / 1 point

☒ We do not need to propagate augmented state vectors and covariance matrices that jointly describe the state, process noise, and sensor noise.

 **Correct**
Yes, this is correct.

☐ In general, we can perform a single Cholesky decomposition per iteration rather than two.

☒ There are fewer evaluations of the state equation per iteration.

 **Correct**
Yes, this is a consequence of x_k being shorter than x_k^a .

☒ There are fewer evaluations of the output equation per iteration.

 **Correct**
Yes, this is a consequence of x_k being shorter than x_k^a .

2. What is a disadvantage of the method presented in this lesson, which is simplified for the case where process and sensor noises are additive?

1 / 1 point

☐ It produces less accurate state estimates in general.

☐ It requires more memory to store intermediate results.

☒ It requires two Cholesky decompositions per iteration instead of only one.

☐ It produces less reliable confidence bounds in general.

 **Correct**
Yes, this is correct.

3. What are some characteristics of the cubature Kalman filter (CKF)?

1 / 1 point

☒ It has no user-specified tuning parameters. That is, γ and $\alpha_i^{(m)}$ and $\alpha_i^{(c)}$ all have a pre-determined form.

 **Correct**
Yes, this is correct.

☒ The tuning parameters $\alpha_i^{(m)}$ and $\alpha_i^{(c)}$ are all non-negative, which makes the filter more robust to roundoff errors.

 **Correct**
Yes, this is true. It helps to preserve the positive-definiteness of the covariance calculations.

☒ It uses one fewer sigma point than the general SPKF.

 **Correct**
Yes, it deletes the sigma point for $i = 0$.

☐ It produces more accurate state estimates than other kinds of SPKF.